

Roll No.

TBT-604

**B. TECH. (BIOTECHNOLOGY)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

NANOBIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss in brief about biological effects of nanoparticles. (CO1)
(b) Describe in brief about practice of microfabrication techniques. (CO1)
(c) Discuss in brief about history of nanobiotechnology. (CO1)
2. (a) Describe top-down and bottom-up approaches. (CO2)
(b) Explain in brief about biological production of metal nano particles. (CO2)
(c) Discuss about nanodevices and its applications used in modern science.
3. (a) Discuss in brief about biomedical research in nanobiotechnology. (CO3)

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(2)

- (b) Discuss Nanodrug delivery and Nanoimaging in detail. (CO3)
- (c) Explain tumor targeting and other diagnostic applications of nanobiotechnology. (CO3)
- 4. (a) Describe biomedical polymers and its applications. (CO4)
- (b) Explain in brief about developing some drug delivery tools through nanobiotechnology. (CO4)
- (c) Describe nanoparticle based immobilization assays. (CO4)
- 5. (a) Describe immuno-nanotechnology and its applications. (CO5)
- (b) Discuss about nanobiotechnology principles used in construction of microelectronic devices. (CO5)
- (c) Describe biosensors in detail. (CO5)

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TBT-603(C)

**B. TECH. (BIOTECHNOLOGY)
(SIXTH SEMESTER)**

END SEMESTER EXAMINATION, 2022

RATIONAL DRUG DISCOVERY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe any *three* types of processes used in Computer aided drug discovery. (CO1)
- (b) What is Homology modelling ? Explain various steps of Homology modelling. (CO1)
- (c) Describe various methods used for drug target identification and validation. (CO1)
2. (a) Differentiate between quantum mechanics and molecular mechanics. (CO2)

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- (b) Explain AMBER and CHARMM. How do we use various types of molecular interactions for calculating force fields ? (CO2)
- (c) Why do we apply energy minimization ? Explain Newton-Raphson algorithm. (CO2)
3. (a) What are the steps of molecular dynamics simulation ? Which command is used to generate topology parameters and add ions ? (CO3)
- (b) What are periodic boundary conditions ? Explain NVT and NPT ensembles and applications. (CO3)
- (c) Explain steps of principal component analysis. How is PCA applied in MD simulation results ? (CO3)
4. (a) Which rule is used to assess ADMET parameters in a potential molecule ? Differentiate between pharmacodynamics and pharmacokinetics. (CO4)
- (b) Explain Zinc and Pub chem. databases used in virtual screening of the drug molecules. (CO4)
- (c) Describe parameters of docking. Explain fragment based and incremental docking algorithms. (CO4)
5. (a) Explain Hansch equation and different molecular descriptors used in QSAR. (CO5)
- (b) Describe Artificial Neural Network and applications of ANN in drug designing. (CO5)
- (c) What is a pharmacophore ? Discuss Pharmacophore modelling in detail. (CO5)

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3/6/2022

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Roll No.

TBT-602

**B. TECH. (BIOTECHNOLOGY)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
PLANT BIOTECHNOLOGY
Time : Three Hours
Maximum Marks : 100**

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Explain in detail the different stages in micropropagation. (CO1)
(b) Define plant tissue culture and discuss its advantages and disadvantages in detail. (CO1)
(c) Explain in detail the technique of virus free plant production. (CO1)
 2. (a) Explain in detail how we can produce *in vitro* plantlets using indirect somatic organogenesis. (CO2)
(b) Explain the protocol for protoplast isolation and discuss the various factors associated with protoplast isolation. (CO2)
(c) Discuss *in vitro* haploid production by *in vitro* pollen culture. (CO2)
 3. (a) What is germplasm ? Explain in detail *in vitro* germplasm conservation via cryopreservation technique. (CO3)

P. T. O.

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- (b) Give a descriptive account of the method, products, and applications of protoplast fusion. (CO3)
- (c) Explain the term bioremediation. Write a detailed note on bioremediation via plants. (CO3)
- 4. (a) What is meant by vector mediated transformation ? Explain in detail pTi based plant transformation. (CO4)
- (b) Write short notes on the following molecular markers : (CO4)
 - (i) RAPD
 - (ii) VNTRs
 - (iii) SNPs
- (c) What is meant by vector less DNA transfer ? Explain in detail the method of particle-mediated gene transfer. (CO4)
- 5. (a) What are genetically modified organisms ? Explain in detail citing appropriate examples. (CO5)
- (b) What is chloroplast engineering ? Explain how can it be superior to nuclear transgenics. (CO5)
- (c) Describe the concept of developing biotic stress-tolerant plants giving appropriate examples. (CO5)

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Roll No.

TBT-602

B. TECH. (BIOTECHNOLOGY)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
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Time : Three Hours
Maximum Marks : 100

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End Semester Examination, 2022

B Tech (Biotechnology)

Semester: Six

COURSE: PYTHON

Course Code: TBT-603b

Time: 3 Hr

Max Marks: 100

Note:

1. All questions are compulsory.
2. Answer any two sub questions among a, b & c in each main question
3. Total marks in each main question are twenty.
4. Each question carries 10 marks

Q 1	Attempt any two parts out of the given three parts.	2 x 10 = 20 Marks	COs
(a)	What are the various data types available in Python? Explain the applications of each type.		CO-1,3
(b)	What are collections in python? Explain each with suitable example.		CO-2,3
(c)	What do you understand by the term decision statement? Explain the python decision statements.		CO-4,5
Q 2	Attempt any two parts out of the given three parts.	2 x 10 = 20 Marks	
(a)	What is a function? How to define a function in python? Write a program using function to find out the given string is palindrome or not.		CO-2,3
(b)	Write a python program to calculate the area of circle, rectangular and triangle. print the results.		CO-3,4
(c)	List out all the useful string methods which supports in python. Explain with an example for each method.		CO-4,5
Q 3	Attempt any two parts out of the given three parts.	2 x 10 = 20 Marks	
(a)	What are the different flow control statements supports in python? Explain any 3 with a suitable example program and flow chart.		CO-2,4
(b)	What is dictionary? Illustrate with an example python program the usage of nested dictionary.		CO-2,3
(c)	List out all the useful string methods which are supported in python. Explain with an example for each method.		CO-1,3
Q 4	Attempt any two parts out of the given three parts.	2 x 10 = 20 Marks	
(a)	What is an array? Explain the concept of array implementations in python with suitable example.		CO-3,5
(b)	What is a module in python? List the useful python modules with some of their functions.		CO-2,4
(c)	What is a class? How to define class in python? How to initiate a class and how the class members are accessed?		CO-4,5
Q 5	Attempt any two parts out of the given three parts.	2 x 10 = 20 Marks	
(a)	What is a Sequence? Explain the methods used concatenating and performing useful functions on sequences.		CO-3,5
(b)	What are Sequence objects in Biopython? Relate the features of Sequences with Strings.		CO-4,5
(c)	Explain the process of Concatenating or adding sequences with real example.		CO-1,3

Roll No.

TBT-601

B. TECH. (BIOTECHNOLOGY)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022

ANIMAL BIOTECHNOLOGY**Time : Three Hours****Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write the name of equipment used in animal cell culture. (CO1)
(b) What is the role of buffer in animal cell culture ? Explain natural buffer. (CO1)
(c) Explain Primary and Secondary cell culture. (CO1)
2. (a) Give an overview on transgenic mice. (CO2)
(b) What is FMD and how does animal biotechnology help in it ? (CO2)
(c) What is cryopreservation ? (CO1)

P. T. O.

(2)

3. (a) Explain embryonic stem cell methods of gene transfer. (CO2)
(b) Give an overview on transgenic pig. (CO2)
(c) What do you understand by oncogenesis ? (CO3)
4. (a) What is gene cloning ? Describe with labelled diagram. (CO4)
(b) What is Transfection ? Explain. (CO3)
(c) Describe the expression system in mammalian cell culture. (CO4)
5. (a) What is ornamental marine species ? (CO5)
(b) What is antisense RNA technology ? (CO4)
(c) What is transgenic fishes ? (CO5)